



Python Programming - 2301CS404

Lab - 10

Roll No : 514

Name : Meet Indoriya

01) WAP to read and display the contents of a text file. (also try to open the file in some other directory)

- in the form of a string
- line by line
- in the form of a list

```
In [13]: fp = open('untitled.txt','r')
data = fp.read()
print(data)
fp.close()

fp = open('untitled.txt','r')
print("\n",fp.readline())
fp.close()

fp = open('untitled.txt','r')
data = fp.readlines()
print(data)
fp.close()
```

```
Hello , Meet
D2D 4-D
514
```

```
Hello , Meet
```

```
['Hello , Meet\n', 'D2D 4-D\n', '514']
```

02) WAP to create file named "new.txt" only if it doesn't exist.

```
In [16]: file1 = open("File1.txt", 'w')
file1.write("New File Named File1.txt")
file1 = open("File1.txt", 'r')
print("File:1 \n", file1.read())
file1.close()
```

File:1
New File Named File1.txt

Out[16]: True

03) WAP to read first 5 lines from the text file.

```
In [24]: f3 = open('untitled.txt')
for i in range(0,5,1):
    print(f3.readline())
f3.close()
```

Hello , Meet

D2D 4-D

514

Darshan University

Computer Science and Engineering

04) WAP to find the longest word(s) in a file.

```
In [113... f4 = open("untitled.txt", 'r')
wordCount2 = 0
data2 = f4.read().split()
print(data2)
f4.close()
```

['Hello', ',', 'Meet', 'D2D', '4-D', '514', 'Darshan', 'University', 'Computer', 'Science', 'and', 'Engineering', 'Computer', 'Science']

05) WAP to count the no. of lines, words and characters in a given text file.

```
In [47]: f5 = open("untitled.txt", 'r')
wordCount = 0
lineCount = 0
characterCount = 0
for i in f5.read():
    characterCount += 1
fp.close()

f5 = open("untitled.txt", 'r')
for i in f5.readlines():
```

```

        lineCount += 1
f5.close()

f5 = open("untitled.txt",'r')
for i in f5.readline():
    wordCount+= 1

print("Character Count:",characterCount)
print("Line Count:",lineCount)
print("Word Count:",wordCount)
f5.close()

```

Character Count: 97
 Line Count: 6
 Word Count: 13

06) WAP to copy the content of a file to the another file.

In [107...

```

f6 = open("untitled.txt",'r')
data = f6.read()
print(data)
newF6 = open("untitled2.txt",'a')
newF6.write(data)
f6.close()
newF6.close()

```

Hello , Meet
 D2D 4-D
 514
 Darshan University
 Computer Science and Engineering
 Computer Science

07) WAP to find the size of the text file.

In [117...

```

f7 = open("untitled.txt", "r")
print(len(f7.read()))
f7.close()

```

93

08) WAP to create an UDF named frequency to count occurrences of the specific word in a given text file.

In [116...

```

def feqCount(string):
    count = 0
    f8 = open("untitled.txt", "r")
    for line in f8:
        word = line.split(" ")
        count += len(word)
    print("Total Number of Words: " + str(count))
    f8.close()

feqCount("Meet")

```

Total Number of Words: 14

09) WAP to get the score of five subjects from the user, store them in a file. Fetch those marks and find the highest score.

```
In [119... sub1 = int(input("Enter marks of the first subject: "))
sub2 = int(input("Enter marks of the second subject: "))
sub3 = int(input("Enter marks of the third subject: "))
sub4 = int(input("Enter marks of the fourth subject: "))
sub5 = int(input("Enter marks of the fifth subject: "))

avg = (sub1 + sub2 + sub3 + sub4 + sub5) / 5

if avg >= 90:
    print("Grade: A")
elif avg >= 80 and avg < 90:
    print("Grade: B")
elif avg >= 70 and avg < 80:
    print("Grade: C")
elif avg >= 60 and avg < 70:
    print("Grade: D")
else:
    print("Grade: F")
f9 = open("Marks.txt", 'a')
avg = str(avg)
f9.write(avg)
print(max(avg))
f9.close()
```

Grade: A

1

10) WAP to write first 100 prime numbers to a file named primenumbers.txt

(Note: each number should be in new line)

```
In [126... def is_prime(n):
    if n < 2:
        return False
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
    return True

count = 0
num = 2
primes = []
while count < 100:
    if is_prime(num):
        primes.append(num)
        count += 1
    num += 1

with open("primenumbers.txt", "w") as f:
    for p in primes:
        f.write(str(p) + "\n")
```

```
print("First 100 prime numbers written to primenumbers.txt")
```

First 100 prime numbers written to primenumbers.txt

11) WAP to merge two files and write it in a new file.

```
In [125... f11 = open("untitled.txt","r")
f11_2 = open("untitled2.txt","r")
f11_3 = open("File1.txt","r")
f11_4 = open("File2.txt","a")
data = f11.read() + f11_2.read() + f11_3.read()
f11_4.write(data)
f11.close()
f11_2.close()
f11_3.close()
f11_4.close()
```

12) WAP to replace word1 by word2 of a text file. Write the updated data to new file.

```
In [129... s = input("Enter text to replace the existing contents:")
f12 = open("New_File.txt", "r+")

f12.truncate(0)
f12.write(s)
f12.close()
print("Text successfully replaced")
```

Text successfully replaced

13) Demonstrate tell() and seek() for all the cases(seek from beginning-end-current position) taking a suitable example of your choice.

```
In [136... f13 = open("untitled.txt","r")
print("Postion Before Reading 5 Character:",f13.tell())
f13.read(5)
print("Postion After Reading 5 Character:",f13.tell())

f13.seek(0,0)
print("Pointer Reset:",f13.read())
f13.close()
```

Postion Before Reading 5 Character: 0

Postion After Reading 5 Character: 5

Poiunter Reset: Hello , Meet

D2D 4-D

514

Darshan University

Computer Science and Engineering

Computer Science

In []: